

Network Device Management

CCNA 200-301 V1.1

MODULE 2

Topics

2.1 Connecting to a Cisco Device

2.2 Basic Cisco Commands

2.3 Configuring for Connectivity

2.4 Neighbor Discovery

2.5 Basic Troubleshooting

2.6 Review



2.1 Connecting to a Cisco Device

CCNA 200-301 v1.1

Module 2

Managing a Cisco device

Access Methods for Cisco Device Management

Console Connection

RADIUS and TACACS+

Platforms for Cloud-Managing Cisco Devices

Managing a Cisco Device

You can manage a Cisco device by directly or remotely connecting a computer to the device

- The computer must be running terminal emulation software

Cisco devices do not have conventional computer ports

- No video, no USB for keyboard or mouse, etc.

To manually configure a Cisco device:

- Connect a rollover (console) cable* from your PC serial port to the device console port
- Make a remote connection to it via SSH or telnet
 - No RDP, VNC, X, etc.
- Some devices provide HTTP/HTTPS services for management via a browser or REST API

A rollover cable is neither a straight through nor a crossover cable.
Instead, it reverses the pins: 8 – 1, 7 – 2, 6 – 3, 5 - 4



Access Methods for Cisco Device Management

Access Method	Description
Console	Connect a rollover cable between a computer serial port and the console port of the Cisco device
Telnet	Make a clear text connection over the network to the device
SSH	Make an encrypted connection over the network to the device
HTTP/HTTPS	Enable the HTTP service on the device, and connect using a browser or REST API
RADIUS	Authentication to the device is redirected across the network to a vendor-neutral RADIUS server, which authenticates the user and determines their privilege level
TACACS+	Authentication to the device is redirected across the network to a Cisco TACACS+ server, which authenticates the user and determines their privilege level
Cloud	Authentication to the device and device management is performed in the cloud



Console Connection

The device will have either (or both) a:

- RJ-45 serial port
- Small USB port for console connections

Configure your terminal emulator settings for:

- Baud Rate (Speed): 9600 bps
- Data Bits: 8
- Parity: None
- Stop Bits: 1
- Flow Control: None

Use a terminal emulator such as PuTTY or Tera Term



RADIUS and TACACS+

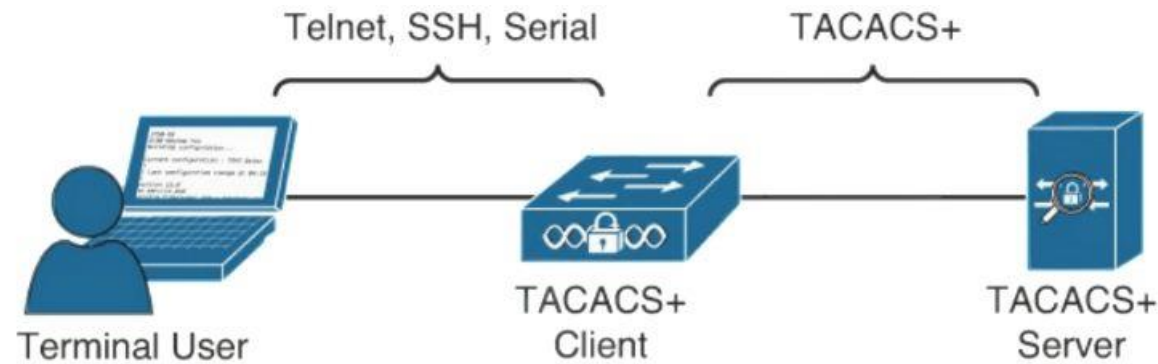
When a user tries to log onto a device, their authentication request is forwarded to a server

The server authenticates the user and sets their privilege level

You must configure both the server and the device (client)

RADIUS is the vendor-neutral implementation

TACACS+ is the Cisco proprietary implementation



Platforms for Cloud-Managing Cisco Devices

Platform	Description
Cisco Meraki Dashboard	Cloud management of Meraki devices, including wireless access points, switches, security appliances
Cisco DNA Center	Centralized control of enterprise networks Can be on-premises or integrated with cloud services
Cisco Cloud Services Router (CSR) 1000V	A virtual router that runs on AWS, Azure, and Google Cloud Used to extend your on-premises network to the cloud
Cisco Smart Accounts and Smart Licensing	Cloud-based management of Cisco device licenses
Cisco SecureX	Security management, threat detection, and response
Cisco SD-WAN (Viptela)	Management, orchestration, and automation of WAN services
Cisco Umbrella	Protects against malicious Internet activity
Cisco WebEx Control Hub	Management platform for Cisco WebEx collaboration devices



2.2 Basic Cisco Commands

CCNA 200-301 v1.1

Module 2

Cisco Command Line Interface (CLI) Privilege Levels

Basic Cisco Commands

Basic Show Commands

Online Help

Abbreviating Commands

Command Modes

The No keyword

Cisco Command Line Interface (CLI) Privilege Levels

There are two basic user privilege levels: User EXEC and Privileged EXEC modes

User EXEC

- Unprivileged mode
- Can perform some show commands
- Perform basic troubleshooting with ping and traceroute
- Can make a telnet or SSH connection to another device
- Prompt is a >
- Can enter privileged mode (if you know the password)

Privileged Exec

- Admin mode
- Can view any information
- Can configure and restart the device
- Prompt is a #



Basic Cisco Commands

Command	Description
<code>enable</code>	Enter Privileged EXEC mode
<code>configure terminal</code>	Enter Global Configuration mode
<code>hostname [name]</code>	Set the device's hostname
<code>enable password</code>	• Require a password to enter privileged mode • Store the password in the configuration file using a reversible encryption method (type 7 encryption)
<code>enable secret</code>	• Store the enable password in the configuration file using the more secure MD5 hash (type 5 encryption) • Overrides any existing enable password • If both an enable password and enable secret are configured on the device, the user must enter the secret
<code>service password-encryption</code>	Encrypt all passwords in the device's configuration file using a type 7 encryption method



Basic Cisco Commands (cont'd)

Command	Description
<code>clock</code>	Set the device clock
<code>banner motd</code>	Set a message of the day that displays as the user logs in
<code>banner login</code>	Set a banner the user sees as before they log in Warn people against unauthorized use
<code>exit</code>	Exit from the current mode to the previous mode
<code>end</code>	Exit from current mode directly to privileged EXEC
<code>copy running-config startup-config</code> <code>write memory</code>	Save the running configuration to NVRAM
<code>reload</code>	Reboot the device
<code>logout</code>	Terminate your session and log out of the device



Basic Show Commands

Command	Description
<code>show version</code>	Display the system hardware and software version
<code>show running-config</code>	Display the current configuration in RAM
<code>show startup-config</code>	Display the configuration stored in NVRAM
<code>show ip interface</code> <code>show ip interface brief</code>	Display a brief summary of IP addresses and interface statuses
<code>show interface</code>	Display all statistics of an interface
<code>show mac address-table</code>	Display the contents of a switch's mac table
<code>show ip route</code>	Display a router's route table
<code>show ip arp</code>	Display a device's arp cache
<code>?</code>	Display online help



Online Help

? after a command will list any subcommands

? (no space) after a few letters lists commands that start with those letters

```
Switch#cl?
```

What commands start with "cl"?

```
clear clock
```

```
Switch#clock ?
```

What are the subcommands of "clock"?

```
set Set the time and date
```

```
Switch#clock
```



Abbreviating Commands

You do not have to type out full words when entering a command at the CLI

You only need to type enough of a word to be unambiguous

```
Switch#s
```

```
% Ambiguous command: "s"
```

```
Switch#sh.
```

```
% Incomplete command.
```

```
Switch#sh?
```

```
show
```

```
Switch#sh ?
```

```
aaa
```

```
access-lists
```

```
arp
```

```
Show AAA values
```

```
List access lists
```

```
Arp table
```

I don't understand which command you mean. There is more than one command that starts with "s"

I understand the command is "show," but you're supposed to include a subcommand



Command Modes

Cisco CLI has different command modes (levels)

The prompt changes depending on which mode you are in

Each mode has its own set of commands

Issue the exit command to leave the mode you are in and go to the mode “below” it

In some cases you can “jump sideways” from one mode to another without having to exit the mode

You can issue the “end” command to jump back to Privileged or User EXEC



Command Modes (cont'd)

Prompt	Mode	Description
R1>	User EXEC mode	Provides access to limited commands
R1#	Privileged EXEC mode	Provides access to all commands
R1 (config) #	Global configuration mode	Make changes to overall device configuration
R1 (config-if) #	Interface configuration mode	Configure settings for a specific interface
R1 (config-subif) #	Sub-interface configuration mode	Configure logical subinterfaces on a physical interface
R1 (config-line) #	Line configuration mode	Configure settings for console, SSH, or Telnet access lines
R1 (config-router) #	Router configuration mode	Configure routing protocols
SW1 (config-vlan) #	VLAN configuration mode	Configure vlans on a switch
rommon>	ROM Monitor (ROMMON) mode	A low-level mode for device recovery, diagnostics, and bootstrap operations



The No Keyword

Used to undo any previous command

Negates any previous configuration

Place the word “no” in front of the command you wish to undo

Example:

```
ip address 192.168.1.1 255.255.255.0
```

```
no ip address 192.168.1.1 255.255.255.0
```



2.3 Configuring for Connectivity

CCNA 200-301 v1.1

Module 2

Configuring Interfaces

Virtual Teletype (VTY) Lines

Telnet

Local User Creation Commands

SSH

Configuring Interfaces

Routers must have their interfaces configured with IP address, subnet mask, and (optional) description

A router interface must be manually enabled to go live

- It remains in an administratively down (shutdown) state until brought up
- Even if administratively brought up, if it is not plugged into the network, it will remain down (line protocol down) until it hears carrier on its receive pins
- You want to see an interface in an “up - up” state

Layer 2 switchports come up immediately when something live is plugged into them

- They just have to detect carrier on the receive pins
- You can manually shut down a switchport

You can configure a switchport by:

- Changing its mode (access, trunk)
- Providing an optional description
- (If possible) Changing it to a Layer 3 port, then adding an IP address and subnet mask



Interface Configuration Commands

Command	Description
<code>interface [type] [number]</code> <code>interface GigabitEthernet0/0</code>	Enter interface configuration mode
<code>ip address [ip address] [subnet mask]</code>	Assign an IP address to a Layer 3 interface
<code>description [text]</code>	Add a description to an interface
<code>no shutdown</code>	Enable the interface (bring it up)
<code>shutdown</code>	Disable the interface (bring it down)



Virtual Teletype (VTY) Lines

A Cisco device can have multiple simultaneous remote administrative connections

Each connection is a “VTY”

Most devices can handle 16 VTY connections at a time (0 – 15)

Older devices can only handle 5 at a time (0 – 4)

You must specify the VTY lines when enabling telnet or SSH connections



Telnet

Old style remote control protocol

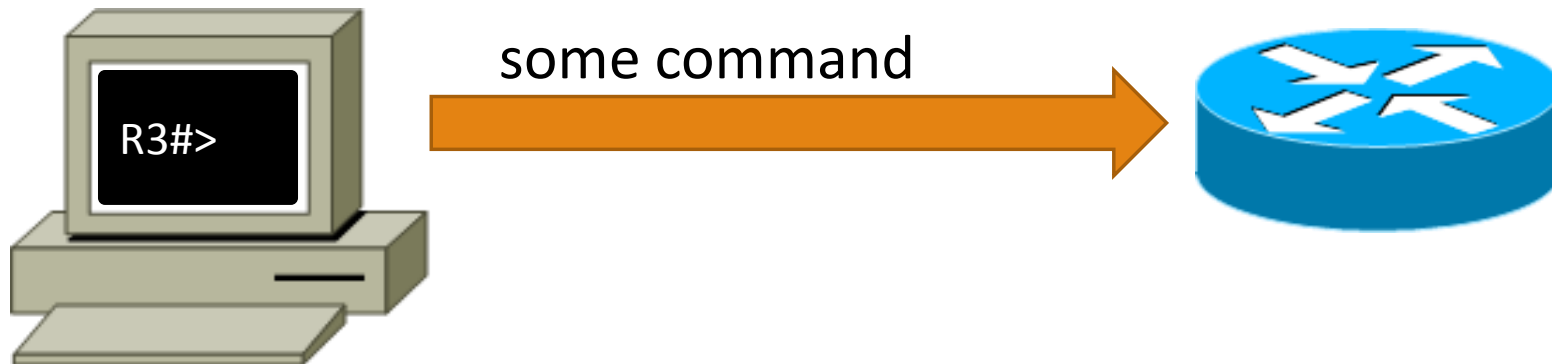
Provides client with a command prompt on a remote device

TCP Port 23

All transmissions are sent and received in clear text

Telnet service automatically installed in Cisco devices (needs to be configured)

Desktop operating systems need to have telnet client installed / enabled



How to Configure a Router for Telnet

High level steps:

1. Set the Router's Hostname
2. Configure an IP Address on an Interface
3. Set Up a Username and Password
4. Enable Telnet on the VTY Lines
5. Require login for VTY Lines
6. Save the Configuration
7. Test the Telnet Connection
8. (Optional) Monitor Telnet Connections



How to Configure a Router for Telnet (cont'd)

```
Router> enable
```

```
Router# configure terminal
```

```
Router(config)# hostname R1
```

```
R1(config)# interface GigabitEthernet0/0
```

```
R1(config-if)# ip address 192.168.1.1 255.255.255.0
```

```
R1(config-if)# no shutdown
```

```
R1(config-if)# exit
```

```
R1(config)# username admin privilege 15 secret LetMeIn
```



How to Configure a Router for Telnet (cont'd)

```
R1(config)# line vty 0 15
```

```
R1(config-line)# login local
```

```
R1(config-line)# transport input telnet
```

```
R1(config-line)# end
```

```
R1# copy running-config startup-config
```

```
R2> telnet 192.168.1.1
```



Local User Creation Commands

Command	Description
<code>username <name> privilege 1 secret <password></code>	• Create a local user account • When the user logs in, they will immediately be given a USER EXEC (unprivileged) prompt
<code>username <name> privilege 15 secret <password></code>	• Create a local user account • When the user logs in, they will immediately be given a PRIVILEGED EXEC prompt



Telnet Commands

Command	Description
<code>line vty 0 4</code> <code>line vty 0 15</code>	Enter vty configuration mode for the desired number of vty telnet sessions (5, or 16)
<code>login local</code>	Username and password is checked against the local database
<code>transport input telnet</code>	Allow telnet connections
<code>telnet <target device IP></code>	Start a telnet session from another Cisco device
<code>CTRL+6+SHIFT, x</code>	Toggle away from the telnet session back to your local CLI
<code>ENTER</code>	Resume your telnet session
<code>exit</code>	End a telnet session
<code>show session</code>	Shows your active outbound telnet sessions to other devices
<code>show users</code>	• Shows active inbound telnet sessions to this device • Shows username, telnet and SSH incoming connections

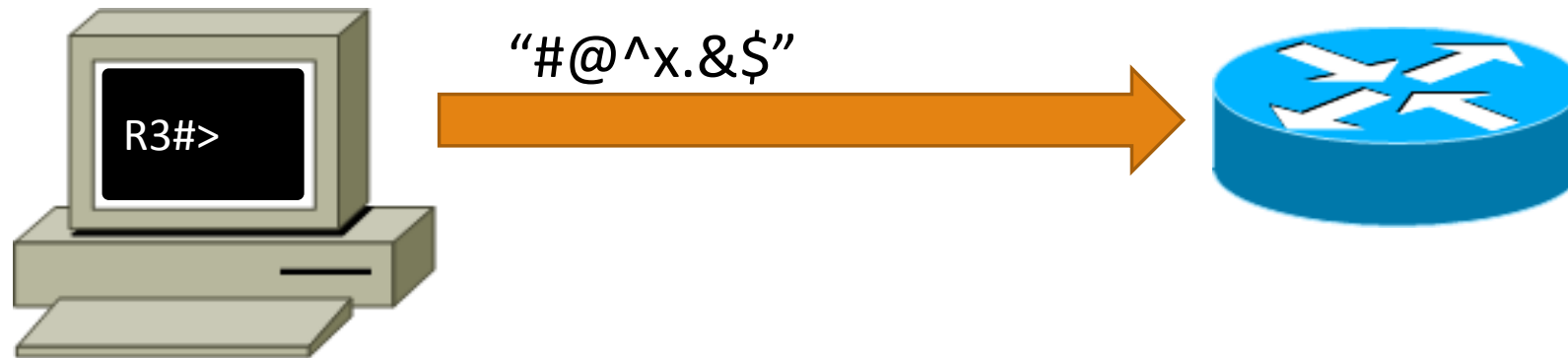


Secure Shell (SSH)

Encrypted replacement for Telnet

Preferred management connection type for Cisco devices

TCP port 22



Secure Shell (SSH)

Both sides trade public keys to create a session key

- Most SSH applications can create their own public/private key pair

SSH Requirements:

- Host name and domain name configured on the device
- Requires k9 (crypto) IOS image on the device

```
Router#show version
```

```
Cisco IOS Software, C2900 Software (C2900-UNIVERSALK9-M), Version 15.1(4)M4
```



How to Configure a Router for SSH

High level steps:

1. Configure the Router's Hostname and Domain Name
2. Generate the SSH Key Pair
3. Configure the SSH Version
4. Create a Local Username and Password
5. Require Authentication
6. (Optional) Set a Time Limit for Successful Authentication and Max Authentication Attempts
7. Save the Configuration
8. Test the SSH Connection



Configuring a Router for SSH

```
Router> enable
```

```
Router# configure terminal
```

```
Router(config)# hostname R1
```

```
R1(config)# ip domain-name mydomain.com
```

```
R1(config)# crypto key generate rsa
```

The name for the keys will be: R1.mydomain.com

Choose the size of the key modulus in the range of 360 to 4096 for your

General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 2048



Configuring a Router for SSH (cont'd)

```
R1(config)# ip ssh version 2
```

```
R1(config)# username admin privilege 15 secret LetMeIn
```

```
R1(config)# line vty 0 15
```

```
R1(config-line)# transport input ssh
```

```
R1(config-line)# login local
```

```
R1(config-line)# exit
```

```
R1(config)# ip ssh time-out 60
```

```
R1(config)# ip ssh authentication-retries 3
```

```
R1(config)# end
```

```
R1# copy running-config startup-config
```



SSH Commands

Command	Description
hostname	Configure a hostname for the device
ip domain-name <domain name>	Provide the device domain name
crypto key generate rsa	When prompted, choose between 360 – 4096 bit key length
ip ssh version 2	Set the SSH version
username <user> privilege 15 secret <password>	Create a local user with a secret password who will have admin privilege in the session



SSH Commands (cont'd)

Command	Description
<code>line vty 0 15</code>	Enter vty configuration mode for 16 vty lines
<code>transport input ssh</code> <code>transport input ssh telnet</code>	Enable SSH only Enable SSH and if that fails, permit telnet
<code>login local</code>	Require local authentication for the user
<code>ip ssh time-out 60</code>	The user must successfully authenticate in 60 seconds
<code>ip ssh authentication-retries 3</code>	The user can retry authentication 3 times before the session is terminated and they must start over



SSH Connection Commands

Command	Description
<code>ssh -l <username> <destination IP></code>	Start an SSH connection from one Cisco device to another
<code>CTRL+SHIFT+6, x</code>	Toggle back to your local CLI
<code>ENTER</code>	Resume your SSH session
<code>exit</code> <code>logout</code>	End your SSH session
<code>show ssh</code>	View active SSH sessions in our out of this device
<code>show users</code>	• Shows active inbound SSH sessions to this device • Shows username, SSH and telnet incoming connections

2.4 Neighbor Discovery

CCNA 200-301 v1.1

Module 2

Cisco Discovery Protocol (CDP)

Link Layer Discovery Protocol (LLDP)

Cisco Discovery Protocol (CDP)

A Cisco proprietary protocol

- Also used by a few other switch, IP phone, and network management vendors

Used to discover information about directly connected devices

- Operates at Layer 2 (Data Link Layer) of the OSI model
- Enables topology discovery
- Verifies Layer 2 connectivity and obtains neighbor IP addresses
- Does not extend past the neighbor – you must connect to the neighbor to see ITS CDP neighbors

It is independent of any Layer 3 protocol

- You do not need an IP address to be configured for it travel across a link
- You DO need Layer 2 connectivity (e.g. Ethernet, PPP, HDLC, etc.) the link must be “up”



CDP Packets

Sent out all physical ports every 60 seconds by default

- Enabled by default on most Cisco switches and routers
- Is not enabled by default on Cisco firewalls

Sent to the multicast address 01-00-0C-CC-CC-CC

Data includes:

- Device identifiers, IP addresses, port identifiers, capabilities (e.g., router, switch), and platform information

Hold Time:

- CDP neighbor information is retained for 180 seconds before it is discarded if no updates are received

Security consideration:

- CDP should be disabled on interfaces that are public facing to avoid information leakage



Enabling CDP

Enable CDP globally:

- `switch(config)# cdp run`
- `switch(config)# no cdp run`

Enable CDP on an interface:

- `switch(config-if)# cdp enable`
- `switch(config-if)# no cdp enable`
- `switch(config-if)# cdp receive`



View CDP Information

```
switch# show cdp
```

```
switch# show cdp interface
```

```
switch# show cdp neighbors
```

```
switch# show cdp neighbors detail
```

```
switch# show cdp entry
```

```
switch# show cdp traffic
```



CDP Commands

Command	Description
<code>cdp run</code>	Enable CDP globally
<code>no cdp run</code>	Disable CDP globally
<code>cdp enable</code>	Enable CDP on a specific interface
<code>no cdp enable</code>	Disable CDP on a specific interface
<code>cdp receive</code>	Only receive CDP packets on an interface, don't transmit any out



Show CDP Commands

Command	Description
<code>show cdp</code>	Displays general CDP information on the device
<code>show cdp interface</code>	Displays the status of CDP on each device interface
<code>show cdp neighbors</code>	Provides a quick overview of connected devices, including their capabilities, platform, and the interfaces
<code>show cdp neighbors detail</code>	Provides detailed information of connected devices including IP address and software version
<code>show cdp entry</code>	Provides detailed information about a specific CDP neighbor
<code>show cdp traffic</code>	Provides statistics on CDP traffic



Link Layer Discovery Protocol (LLDP)

Vendor neutral alternative to CDP

Enables network topology mapping and device communication

Information includes chassis ID, port ID, system name, capabilities, and other generic data

Ideal for environments with network hardware from multiple vendors

Preferred in situations where adherence to open standards is required or where interoperability is critical

Often used with Cisco IP phones for VLAN assignment and other configurations



Configuring LLDP

```
SW1# configure terminal
SW1(config)# lldp run
SW1(config)# no lldp run
SW1(config)# lldp reinit 5
SW1(config)# interface [interface-id]
  ◦ Example: int g0/1
```



Configuring LLDP

```
SW1(config-if)# lldp transmit  
SW1(config-if)# lldp receive  
SW1(config-if)# no lldp transmit  
SW1(config-if)# no lldp receive  
SW1(config)# end  
SW1# copy running-config startup-config
```



View LLDP Information

```
SW1# show lldp
```

```
SW1# show lldp neighbors
```

```
SW1# show lldp neighbors detail
```



LLDP Commands

Command	Description
<code>lldp run</code>	Enable LLDP globally
<code>no lldp run</code>	Disable LLDP globally
<code>lldp reinit <seconds></code>	• If LLDP configuration changes, wait x seconds before reinitializing the LLDP process • Helps avoid network instability caused by frequent or rapid changes
<code>lldp transmit</code>	Enable sending LLDP on a specific interface
<code>no lldp transmit</code>	Disable sending LLDP on a specific interface
<code>lldp receive</code>	Enable receiving LLDP on a specific interface
<code>no lldp receive</code>	Disable receiving LLDP on a specific interface



Show LLDP Commands

Command	Description
<code>show lldp</code>	View global LLDP statistics
<code>show lldp neighbors</code>	View LLDP neighbors including name, ports and MAC address
<code>show lldp neighbors detail</code>	Provides detailed information about the neighbor including system capabilities and IP address

2.5 Basic Troubleshooting

CCNA 200-301 v1.1

Module 2

Uptime / Downtime

Cable Issues

Show Interface

Link Up or Down

Line Protocol Down Causes

Network Metrics

Speed Mismatch

Duplex Mismatch

Errors

Real-time Traffic Metrics

Basic Troubleshooting Commands

Uptime / Downtime

The most basic metric you can track on a device, link or system

How long something has been down or up is a very common starting point for other investigations

You can use outside systems to regularly ping a device or service and log any failures to respond

You can also check timestamps in a system log to see time gaps, as well as when a system or service restarted

Use the **show version** command to view device uptime



Cable Issues

Cable too long – exceeds the recommended length for speed or use case

Incorrect cable type – cable has the wrong specification for the use case

UTP/STP cable broken

- One or more of the wires in twisted pair cable is broken
- If you see a link light, that only guarantees that you hear carrier, not that signal is clean or you can transmit

Bad plug/port

- Dirty/corroded/broken pins
- One or more of the wires came loose inside the plug

EMI / RFI interference

- Insufficient shielding or twists to resist the amount of electromagnetic/radio interference in the environment

Fiber optic cable light leakage

- Cable has a bend that exceeds its specified bend radius



Show Interface

Link State (Up / Down)

L2 & L3 Addresses

BW, Reliability, Tx and Rx load

Speed / Duplex

Encapsulation type

```
SW1# show interface
```

```
SW1# show interface g0/1
```



Show Interface Example

```
Router#show int g0/0
GigabitEthernet0/0 is up, line protocol is up (connected)
  Hardware is CN Gigabit Ethernet, address is 00e0.b01d.8b01
  Internet address is 192.168.1.1/24
  MTU 1500 bytes, BW 1000000 Kbit, DLY 10 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Full-duplex, 100Mb/s, media type is RJ45
```



Link Up or Down

Link is up / up:

```
Router#show int g0/0
```

```
GigabitEthernet0/0 is up, line protocol is up (connected)
```



Link Up or Down

Link is up / up:

```
Router#show int g0/0  
GigabitEthernet0/0 is up, line protocol is up (connected)
```

Admin enabled the link, but there is something wrong at Layer 1 or 2:

```
GigabitEthernet0/0 is up, line protocol is down (disabled)
```



Link Up or Down

Link is up / up:

```
Router#show int g0/0  
GigabitEthernet0/0 is up, line protocol is up (connected)
```

Admin enabled the link, but there is something wrong at Layer 1 or 2:

```
GigabitEthernet0/0 is up, line protocol is down (disabled)
```

Admin shut the interface, making it down / down:

```
GigabitEthernet0/0 is administratively down, line protocol is down (disabled)
```



Line Protocol Down Causes

Administrative shutdown

Cable disconnection or failure

Poor line/signal quality

Speed or Duplex mismatch

EtherChannel negotiation failure

Encapsulation (Layer 2 protocol) mismatch

VLAN mismatch

STP port blocking

PoE overload

- End device pulls more power than switchport can deliver



Line Protocol Down Example

```
Router#show ip int brief
```

Interface	IP-Address	OK?	Method	Status
GigabitEthernet0/0	192.168.1.1	YES	manual	up
GigabitEthernet0/1	unassigned	YES	unset	administratively down
GigabitEthernet0/2	unassigned	YES	unset	administratively down
Vlan1	unassigned	YES	unset	administratively down



Network Metrics

Maximum Transmission Unit (MTU)

- Max allowed frame size in bytes -typically 1500, but can be lower for PPP dialup (576), or VPNs (1400+)

Bandwidth (BW)

- Maximum link speed
- Used by routing protocols to determine link speed

```
BW 1000000 Kbit/sec
```

Look farther down to see actual current link speed

```
Full-duplex, 1000Mb/s,
```



Network Metrics (cont'd)

Bandwidth utilization

- The percentage of bandwidth being utilized
- Expressed as a fraction, with values ranging from 1/255 (almost no load) to 255/255 (100% utilization)
- txload – current outgoing utilization
- rxload – current incoming utilization

Packet drops

- The number of packets on a network that do not reach their intended destination

Packet error rate

- The frequency of errors



Speed Mismatch

In most cases, there will be no connection

- Interface will show line protocol down

If there is a connection, it would be unstable with high error rates

- The faster device may overwhelm the slower device's buffers
- This leads to dropped packets and significant degradation in network performance

Sometimes, devices may fall back to the lowest common speed (e.g., 10 Mbps) when they cannot agree on a higher speed



Duplex Mismatch

Results in interface errors: Collisions, Late collisions

One Device in Full-Duplex and the Other in Half-Duplex:

The full-duplex device:

- may transmit data while the half-duplex device is also trying to transmit, leading to a collision
- Since the full-duplex device is not expecting any collisions, it continues transmitting without detecting the collision

The half-duplex device:

- which does detect the collision, stops transmitting and waits for a random backoff period before trying again
- Because the collision is detected late (after the first 64 bytes of the frame), it is classified as a **late collision**



Interface Errors

Input / Output rates	5 minute input rate 0 bits/sec, 0 packets/sec
	5 minute output rate 0 bits/sec, 0 packets/sec
Runts	956 packets input, 193351 bytes, 0 no buffer
Giants	Received 956 broadcasts, 0 runts, 0 giants, 0 throttles
	0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
Input / Output errors	0 watchdog, 0 multicast, 0 pause input
	0 input packets with dribble condition detected
CRC Errors	2357 packets output, 263570 bytes, 0 underruns
	0 output errors, 0 collisions, 10 interface resets
Collisions	0 babbles, 0 late collision, 0 deferred
	0 lost carrier, 0 no carrier
Late collision	0 output buffer failures, 0 output buffers swapped out



Frame Errors

Runts

- Ethernet frames that are smaller than the minimum allowed size of 64 bytes
- Have a malformed or incomplete payload
- Caused by duplex mismatches
- Or collisions due to network congestion – node that detects collision abruptly stops transmitting

Giants

- Frames that exceed the MTU
- Caused by MTU mismatch (some use cases use larger frame sizes to carry data more efficiently)
- Can also be caused by collisions – a partial frame “attaches” itself to a normal frame

Increasing input errors and CRC counters:

- Indicate corrupted incoming frames (Frame Check Sequence Failures)



Collision Errors

Collisions:

- Transmitted frame collides with another
- Indicate the link is on a shared Ethernet segment
- Another device is trying to transmit at the same time

Late Collisions:

- Mismatched duplex or cable too long
- Late collision counter increments after 64 bytes are received

Encapsulation Errors

- Frames that are not the expected Ethernet (or other L2 protocol) type

Input Errors: Look at CRC, frame, overrun, and ignored errors

Output Errors: Look at outgoing packet underruns and collisions

Throughput Issues:

- Caused by physical cable faults (indicated by high FCS-Err count)



Real-time Traffic Metrics

Monitor Quality-of-Service (QoS) for real-time traffic such as VoIP, collaborative video, online gaming

Packet loss

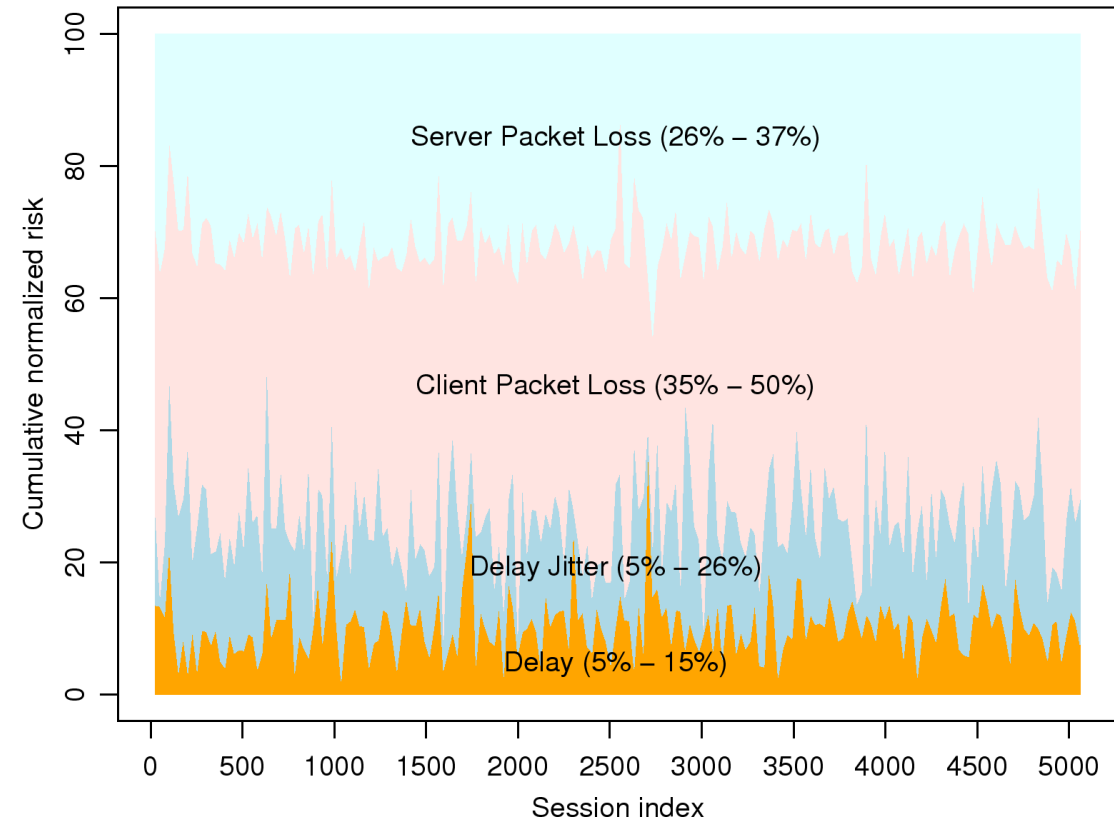
- Packets that never arrive

Latency

- Delay between transmission and reception

Jitter

- Variable delay
- Worst impact is on audio
- Hardest to compensate for



Basic Troubleshooting Commands

Command	Description
<code>show version</code>	Provides detailed information about the device's software and hardware including the IOS version, image file, device model, uptime, memory, and configuration register
<code>show interface</code>	View Layer 2 status of an interface
<code>show ip interface</code> <code>show ip interface brief</code>	Display detailed interface IP information Display status of all interfaces in table format
<code>ping [destination]</code>	Test Layer 3 connectivity to a remote host (ICMP echo, echo reply)
<code>ping</code>	• Advanced ping • Specify source, datagram size, repeat count, timeout and more
<code>traceroute [destination]</code>	• Trace the path to a remote host • Manipulates the TTL of a UDP packet to collect ICMP expired in transit messages from hops along the path
<code>show controller</code>	View Layer 1 status of an interface



Review

2.6 Review

CCNA 200-301 v1.1

Module 2

Review

There are a number of ways in which you can manage a Cisco device

The most direct way is to connect a rollover cable from the serial Port of your PC to the console Port of the Cisco device

You can also make remote connections using telnet, SSH, and in some cases http

You can use RADIUS or TACACS+ to redirect authentication to a central server when a user tries to log on (either locally or remotely) to a device

There are also cloud-based management platforms for Cisco devices



Review (cont'd)

The Cisco command line interface (CLI) has two basic privilege levels: User EXEC and Privileged EXEC

User EXEC is an unprivileged mode

- It allows you to make telnet or SSH connections to another device, view some information and perform basic troubleshooting

When you are in User EXEC mode, your prompt is a greater than sign (>)

You can enter privileged mode from user exec if you know the password

Privileged mode is an administrative mode that allows you to view or configure anything in the device

When you are in privileged mode, your prompt is a pound sign (#)

Use the enable command to enter privileged exec mode

Use the configure terminal command to enter Global configuration mode



Review (cont'd)

Use the `enable password` or `enable secret` command to set a password for entering privileged exec mode

The `enable password` command stores the password in reversible encryption (type 7)

The `enable secret` command stores the password as an md5 hash

If you have both an `enable password` and `enable secret` configured, the user will be required to enter the `enable secret`

The `service password-encryption` command will encrypt all existing passwords in the configuration using a type 7 reversible encryption method



Review (cont'd)

The clock command configures the device clock

The banner command allows you to set a login Banner or a message of the day that displays as the user logs in

You save your configurations by entering the command `copy running-config startup-config`

There are many show commands

- You can only run them at either the top of privileged exec or user exec

The Cisco CLI provides online help in the form of a ?

If you type a? immediately after a few letters, it will return any commands that begin with those letters

If you type a command followed by a space and then a ? it will return all subcommands for that command



Review (cont'd)

In Cisco, you can abbreviate commands by just typing enough of the command to be unambiguous

If your command is ambiguous, you will get an error message stating such

If your command is unambiguous, but expects a subcommand, you will get an error message of incomplete command

The no keyword will negate any command that was previously issued

Type the same command you wish to undo with the word no in front of it



Review (cont'd)

In addition to user exec and privileged exec, the Cisco CLI has various command modes

Each mode has its own set of commands

You enter the basic command for that mode to enter that mode and your prompt changes

You can type exit to leave that mode and go to the mode below it

You can sometimes move sideways from one mode to another without having to exit to a lower level

If you issue the end command, it will jump you back to either user exec or privileged exec depending upon your privilege level



Review (cont'd)

A router interface must be manually configured and enabled before it comes up

A layer 2 switch port will immediately come up if something live is plugged into it

You issue the `no shutdown` command to enable an interface

You issue the `shutdown` command to disable an interface



Review (cont'd)

The high level steps for configuring a device for telnet are:

1. Set the Router's Hostname
2. Configure an IP Address on an Interface
3. Set Up a Username and Password
4. Enable Telnet on the VTY Lines
5. Require login for VTY Lines
6. Save the Configuration
7. Test the Telnet Connection
8. (Optional) Monitor Telnet Connections



Review (cont'd)

The high level steps for configuring a device for SSH are:

1. Configure the Router's Hostname and Domain Name
2. Generate the SSH Key Pair
3. Configure the SSH Version
4. Create a Local Username and Password
5. Require Authentication
6. (Optional) Set a Time Limit for Successful Authentication and Max Authentication Attempts
7. Save the Configuration
8. Test the SSH Connection
9. (Optional) Monitor SSH Connections



Review (cont'd)

Cisco Discovery Protocol (CDP) is a Cisco proprietary Layer 2 protocol that is used to discover immediately connected neighbors

- It is not dependent on layer 3 connectivity
- It only needs for the link to be up to work

You can enable and disable CDP at both the global level as well as the interface level

For security, consider disabling CDP on any interfaces that are public-facing



Review (cont'd)

CDP mostly works on only Cisco devices, though there are a few other vendors that support CDP

So long as the link is up, Cisco devices will regularly send out CDP packets announcing themselves to their immediate neighbors

You can use the `show cdp neighbors` command to discover neighbors including detailed information such as their IP address

A CDP packet will not travel past the immediate neighbor

- If you wish to know what devices are connected to the neighbor, you must connect to that neighbor and issue the `show cdp neighbors` command on it



Review (cont'd)

Link layer Discovery protocol (LLDP) is a vendor neutral alternative to CDP

Use LLDP when working with hardware for multiple vendors including IP phones

You can use the `show version` command to view device uptime, platform, operating system, Hardware, and configuration register

When troubleshooting cable issues, you could have a cable that:

- exceeds the recommended length for the use case
- is the incorrect cable type for the use case
- does not have sufficient shielding or twists for the level of EMI / RFI interference
- Has a bad plug or port with dirty, broken or corroded pins

If the cable is fiber optic, it might have light leakage due to a bend in the cable that exceeds its specified bend radius



Review (cont'd)

The `show interface` command will show:

- If a link is up or down
- The MAC address
- The IP address (if applicable)
- The Maximum Transmission Unit (MTU)
- Available Bandwidth (BW)
- Link reliability
- Transmit and receive loads
- Duplex and actual speed
- Encapsulation type

When you administratively shut an interface, the line protocol also goes down



Review (cont'd)

Other causes of line protocol to be down include:

- bad cable
- poor signal quality
- speed or duplex mismatch
- Encapsulation (Layer 2 protocol) mismatch
- VLAN mismatch
- EtherChannel negotiation failure
- Power over Ethernet overload on the switchport
- Spanning-tree port blocking

When you issue the `show interface` command, and you see collisions and late collisions, this is often caused by duplex mismatch



Review (cont'd)

Runts are frames that are too small to meet the minimum size requirement for the Layer 2 protocol

- They are caused when a collision in Ethernet causes a transmitting node to abruptly stop transmitting a frame

Giants are frames that are too big for the maximum transmission unit size

- They can be too big on purpose (when the switch is specially configured for larger frames—such as in storage communications)
- or when there's a collision and a runt attaches itself to a normal frame

When you see increasing input errors and increasing CRC counters, this indicates corrupted incoming frames

- These frames are failing their check sequences

Collisions occur normally in Ethernet, but only on a shared segment

- They occur when two nodes are trying to transmit at the same time

Late collisions occur after the first 64 bytes of a frame has been transmitted

- It typically happens because there is a duplex mismatch or the cable is too long



Review (cont'd)

It is possible to have encapsulation errors when two nodes are configured for different Layer 2 protocols

When you have real time traffic such as voice over IP, collaborative video, and online gaming, you are concerned with three basic traffic metrics:

- Packet loss
- Latency
- Jitter (variable delay)



Review (cont'd)

Basic troubleshooting commands include:

- `show interface`
- `show ip interface`
- `show ip interface brief`
- `ping [destination]`
- `ping`
- `traceroute [destination]`
- `show controller`